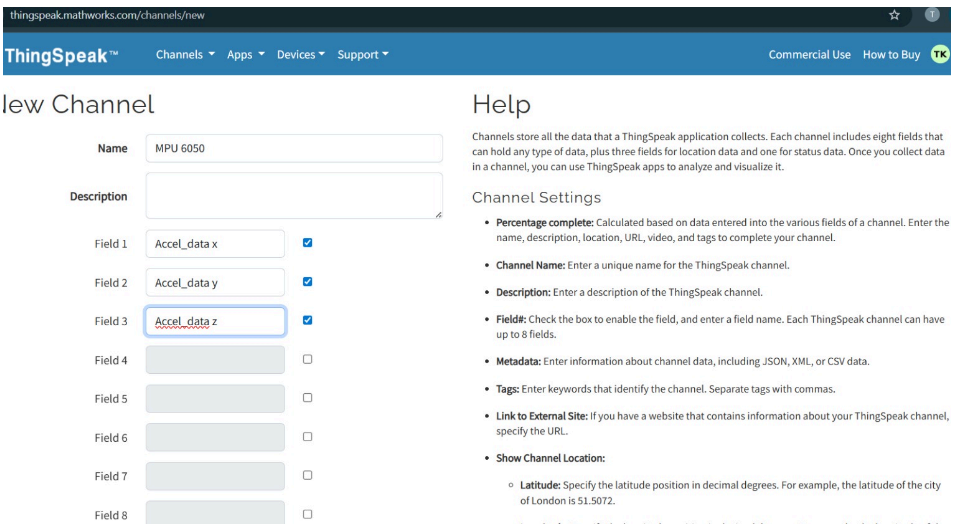
Practical No.9

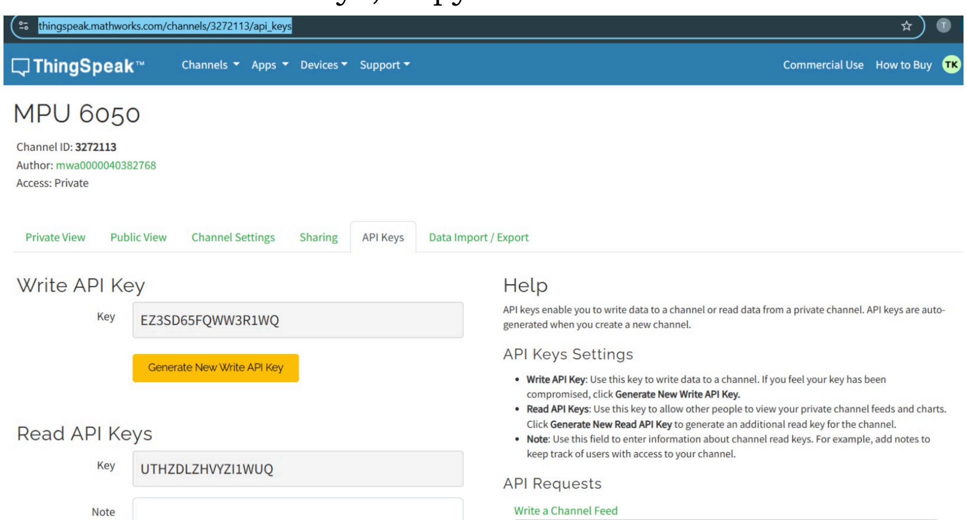
Aim:- To send sensor data online to a cloud server using ThingSpeak and Raspberry Pi.

Steps:-

1. Create an account on thingspeak.
2. Create a New Channel.
3. Fill in the required details:-
   Name = Name of the sensor (e.g., Temperature Sensor, etc).
   Fields = Type of the Data(e.g., Temperature in
4. Save the Channel.



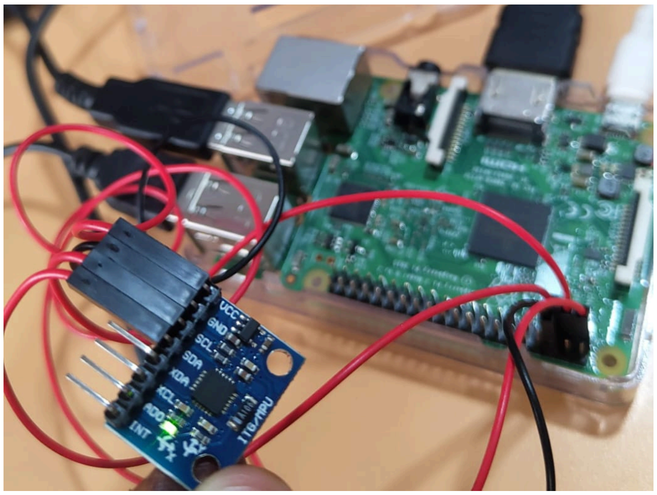
5. Click on API Keys, Copy Write API KEY.

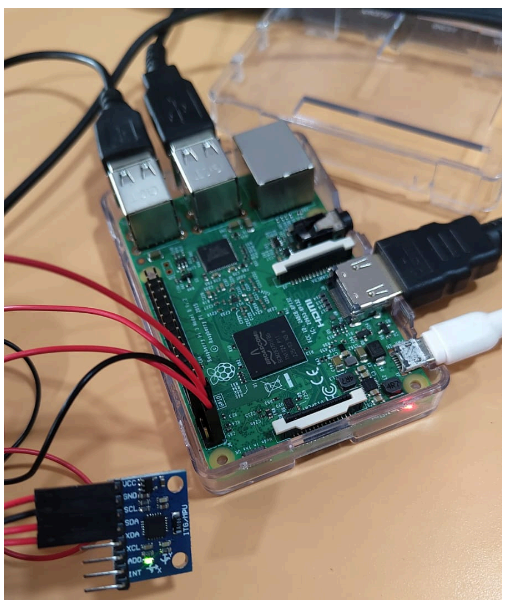


6. Connect sensor MPU-6050 Module 3 Axis Gyroscope Accelerometer Module and RaspberryPi.

The MPU-6050 module typically uses four pins for a basic connection to the Raspberry Pi's GPIO header.

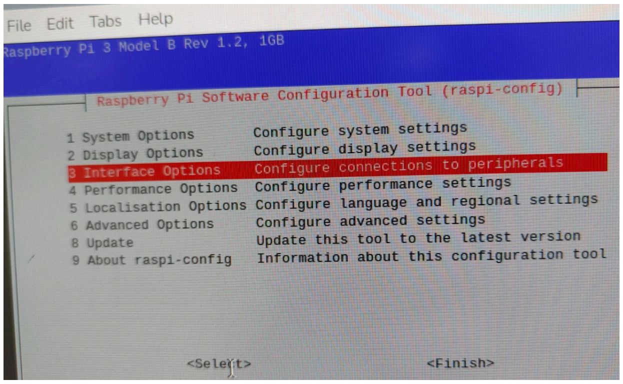
- **VCC**: Connect to Raspberry Pi **3.3V (Pin 1)**.
- **GND**: Connect to Raspberry Pi **Ground (Pin 6 or 9)**.
- **SDA**: Connect to Raspberry Pi **SDA (Pin 3, GPIO 2)**.
- **SCL**: Connect to Raspberry Pi **SCL (Pin 5, GPIO 3)**.



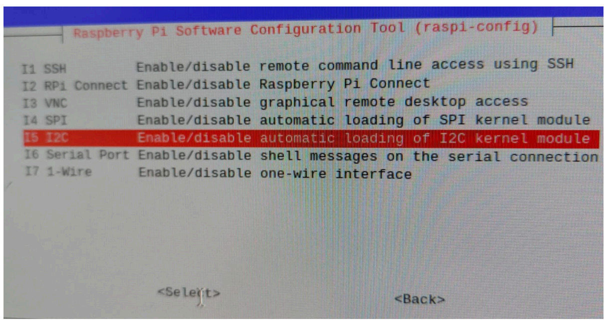


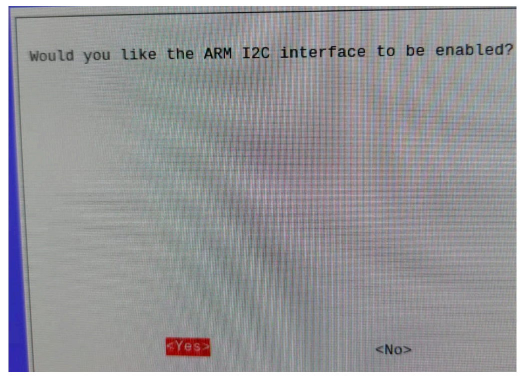
7. Enable I2C in Raspberry Pi using:

   sudo raspi-config

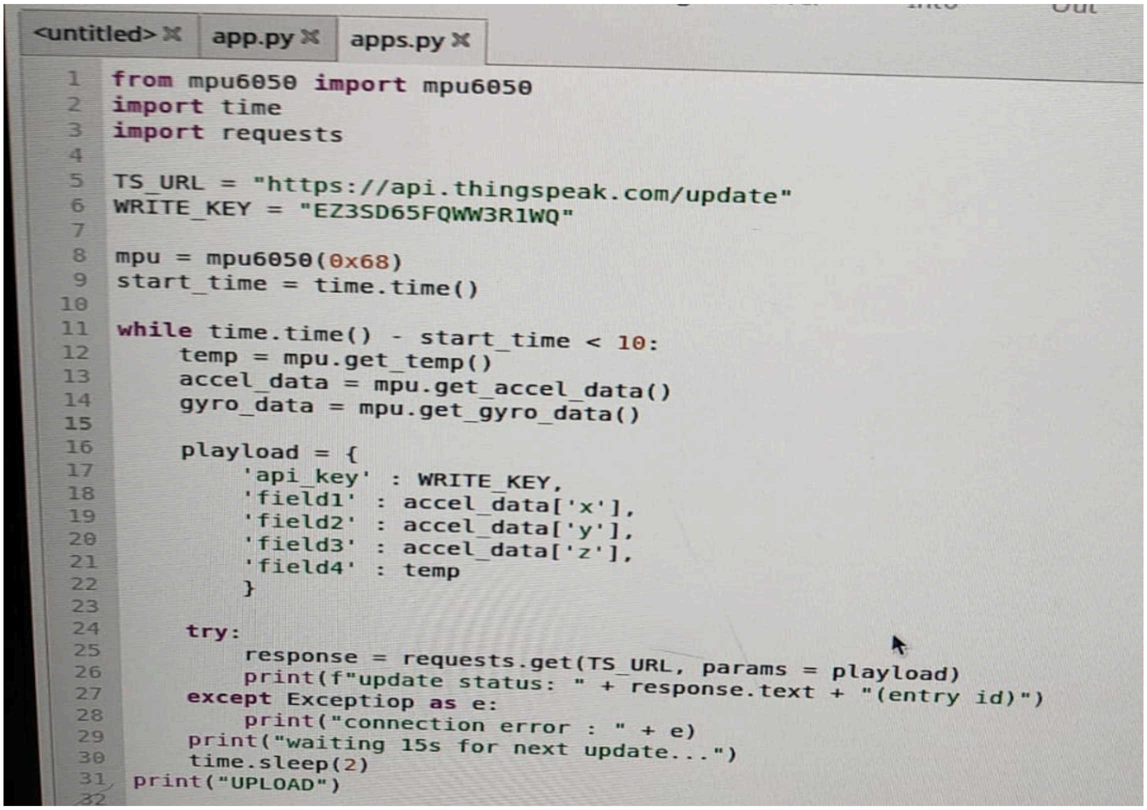


Then enable I2C from Interface Option



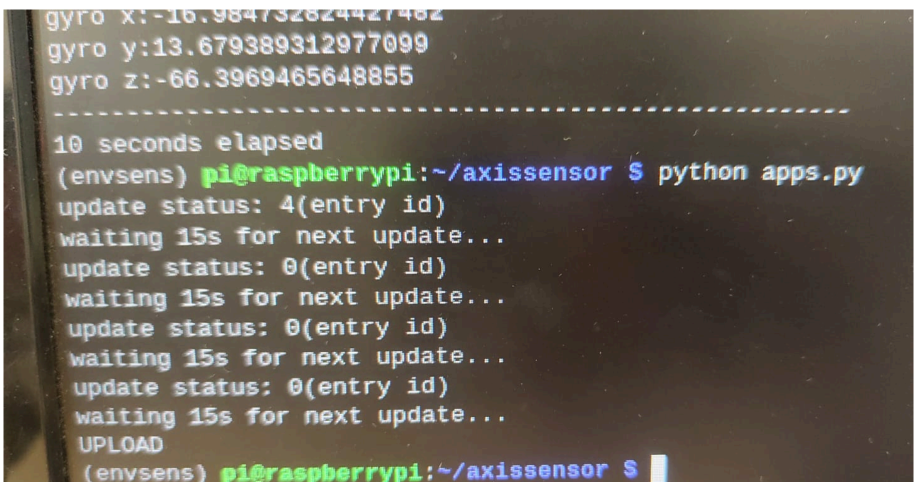


8. Create Folder:-
   mkdir mpu_project
   cd mpu_project
9. Create Python Environment:-
   python3 -m venv myenv.
   source myenv/bin/activate
   Install libraries: mpu6050, requests.
   pip install mpu6050-raspberrypi requests
10. Write Python code:- And Paste API key in WRITE_API variable.



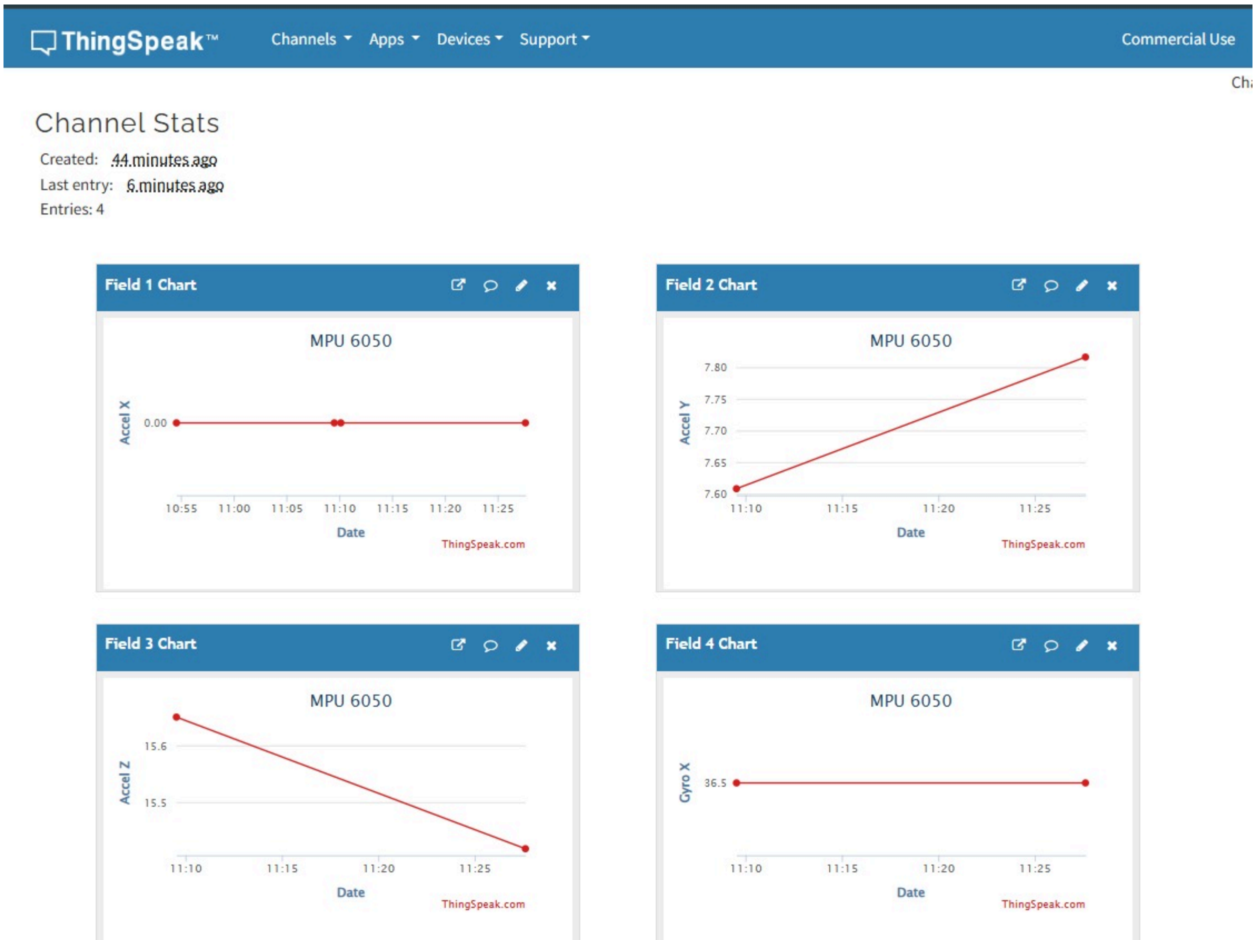
```python
from mpu6050 import mpu6050
import time
import requests

TS_URL = "https://api.thingspeak.com/update"
WRITE_KEY = "EZ3SD65FQWW3R1WQ"

mpu = mpu6050(0x68)
start_time = time.time()

while time.time() - start_time < 10:
    temp = mpu.get_temp()
    accel_data = mpu.get_accel_data()
    gyro_data = mpu.get_gyro_data()

    playload = {
        'api_key' : WRITE_KEY,
        'field1' : accel_data['x'],
        'field2' : accel_data['y'],
        'field3' : accel_data['z'],
        'field4' : temp
        }

    try:
        response = requests.get(TS_URL, params = playload)
        print(f"update status: " + response.text + "(entry id)")
    except Exceptiop as e:
        print("connection error : " + e)
    print("waiting 15s for next update...")
    time.sleep(2)
print("UPLOAD")
```

Output:-



```
gyro y:13.679389312977099
gyro z:-66.3969465648855
--------------------------------------------------
10 seconds elapsed
(envsens) pi@raspberrypi:~/axissensor $ python apps.py
update status: 4(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
update status: 0(entry id)
waiting 15s for next update...
UPLOAD
(envsens) pi@raspberrypi:~/axissensor $
```

Reading Graph on ThingSpeak


ThingSpeak™
Channels
Apps
Devices
Support
Commercial Use
Channel Stats
Created: 44 minutes ago
Last entry: 6 minutes ago
Entries: 4
Field 1 Chart
MPU 6050
Accel X
0.00
10:55
11:00
11:05
11:10
11:15
11:20
11:25
Date
ThingSpeak.com
Field 2 Chart
MPU 6050
Accel Y
7.80
7.75
7.70
7.65
7.60
11:10
11:15
11:20
11:25
Date
ThingSpeak.com
Field 3 Chart
MPU 6050
Accel Z
15.6
15.5
11:10
11:15
11:20
11:25
Date
ThingSpeak.com
Field 4 Chart
MPU 6050
Gyro X
36.5
11:10
11:15
11:20
11:25
Date
ThingSpeak.com